Algebraic fraction. Cancelling fractions

The fundamental property of a fraction, and the single rule that governs every simplification in this chapter: factorise first, cancel second.

LESSON 6–8 | 3 × 40 MIN | ALGEBRA 8, §2 | STAGE 9 · 2.5

LESSON CLOCK

5' | 12' | 8' | 13' | 2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- State what an algebraic fraction is and give its permissible values.
- Use the fundamental property of a fraction in both directions.
- Cancel a fraction by factorising the numerator and the denominator.
- Handle the sign rules, including the swap $b - a = -(a - b)$.

TEXTBOOK

National	§2, pp. 12–17
Cambridge	Learner's Book pp. 40–44
Workbook	Workbook 2.5

01 Explanation

What an algebraic fraction is

DEFINITION

An expression of the form $\frac{A}{B}$, where A and B are polynomials and B is not the zero polynomial, is called an **algebraic fraction**. A is the numerator, B the denominator.

It is defined only for values that keep $B \neq 0$. Every answer in this chapter carries that condition, whether or not the question asks for it.

The fundamental property

THE RULE EVERYTHING RESTS ON

$$\frac{A}{B} = \frac{A \cdot C}{B \cdot C}, \quad \text{where } C \neq 0$$

Multiplying — or dividing — the numerator and the denominator by the same non-zero expression does not change the fraction.

Read left to right it lets you **build up** a fraction to a required denominator, which is what you need for addition. Read right to left it lets you **cancel**, which is what you need now. It is the same rule both ways.

Cancelling — the method

1. **Factorise** the numerator completely.
2. **Factorise** the denominator completely.
3. Cancel every factor common to both.
4. State the permissible values.

$$\frac{x^2 - 9}{x + 3} = \frac{(x-3)(x+3)}{x+3} = x - 3, \; x \neq -3$$

THE ONE MISTAKE THAT COSTS THE MOST MARKS

You may cancel **factors**, never **terms**. $\frac{x+3}{3}$ is **not** x — the 3 in the numerator is added, not multiplied. If you cannot see a bracket around the whole numerator, you cannot cancel.

Signs, and the swap that saves you

A minus sign may sit in three places, and all three mean the same thing:

$$-\frac{a}{b} = \frac{-a}{b} = \frac{a}{-b}$$

The move you will use constantly:

$$b - a = -(a - b)$$

So $\frac{3a - 3b}{b - a} = \frac{3(a-b)}{-(a-b)} = -3$. Whenever a bracket appears reversed, pull out a minus sign and the factors match.

02 Worked examples

EXAMPLE 1 ***Cancel*** $\mathbf{\dfrac{x^2 - 6x + 9}{x^2 - 9}}$

1. $x^2 - 6x + 9 = (x - 3)^2$

   Numerator: perfect square.

2. $x^2 - 9 = (x - 3)(x + 3)$

   Denominator: difference of two squares.

3. $\dfrac{(x - 3)(x - 3)}{(x - 3)(x + 3)}$

   Now the common factor is visible.

4. $\dfrac{x - 3}{x + 3}$

   Cancel one (x − 3) from top and bottom.

5. $x \neq 3, x \neq -3$

   Both zeros of the original denominator stay excluded.

ANSWER

$\dfrac{x - 3}{x + 3}, \; x \neq \pm 3$

EXAMPLE 2 ***Cancel*** $\dfrac{2x^2 + 7x + 3}{x^2 - 9}$

1. $2 \cdot 3 = 6$, and $1 + 6 = 7$

   Split the middle term of the numerator.

2. $2x^2 + x + 6x + 3 = x(2x + 1) + 3(2x + 1) = (2x + 1)(x + 3)$

   Group and factor.

3. $x^2 - 9 = (x - 3)(x + 3)$

   Denominator.

4. $\dfrac{(2x + 1)(x + 3)}{(x - 3)(x + 3)} = \dfrac{2x + 1}{x - 3}$

   Cancel (x + 3).

ANSWER

$\dfrac{2x + 1}{x - 3}, \; x \neq \pm 3$

EXAMPLE 3 **Cancel** $\dfrac{m^2 - mn}{n^2 - mn}$

1. $m^2 - mn = m(m - n)$

   Common factor m.

2. $n^2 - mn = n(n - m)$

   Common factor n.

3. $n - m = -(m - n)$

   Reverse the bracket so it matches the numerator.

4. $\dfrac{m(m - n)}{-n(m - n)} = -\dfrac{m}{n}$

   Cancel (m − n); the minus sign stays.

ANSWER

$-\dfrac{m}{n}$, $n \neq 0, m \neq n$

03 Interactive model

Do the first one with the class; let a learner predict each next step before you reveal it.

Cancelling step by step

$$\frac{x^2 - 9}{x + 3}$$

1. $x^2 - 9 = (x - 3)(x + 3)$

Difference of two squares.

2. $$\frac{(x - 3)(x + 3)}{x + 3}$$

The denominator is already a single factor.

3. $x - 3$

Cancel (x + 3) — but remember x ≠ −3.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	OʻZBEKCHA	РУССКИЙ
Algebraic fraction	Algebraik kasr	Алгебраическая дробь
Fundamental property of a fraction	Kasrning asosiy xossasi	Основное свойство дроби
Cancel (reduce) a fraction	Kasrni qisqartirish	Сокращение дроби
Factor	Koʻpaytuvchi	Множитель
Term	Had	Слагаемое
Polynomial	Koʻphad	Многочлен
Identical transformation	Ayniy almashtirish	Тождественное преобразование
Sign	Ishora	Знак
Bracket	Qavs	Скобка
Simplify	Soddalashtirish	Упростить

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$\frac{x + 3}{3}$ simplifies to:

- x
- $x + 1$
- it does not simplify
- $\frac{x}{3}$

$\frac{6a}{9a}$ equals:

- $\frac{2}{3}$
- $\frac{2a}{3}$
- $\frac{6}{9a}$
- 3

$\frac{x^2 - 4x}{x - 4}$ equals:

- $x - 4$
- x
- $4x$
- $\frac{x}{4}$

$b - a$ is the same as:

- $a - b$
- $-(a - b)$
- $-(b - a)$
- $(a - b)^2$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *Cancel:* $\frac{6a}{9a}$

ANSWER $\frac{2}{3}$, $a \neq 0$

2. *Cancel:* $\frac{12x^2}{4x}$

ANSWER $3x$, $x \neq 0$

3. *Cancel:* $\frac{5ab}{10a}$

ANSWER $\frac{b}{2}$, $a \neq 0$

4. *Cancel:* $\frac{x^2}{x^5}$

ANSWER $\frac{1}{x^3}$, $x \neq 0$

5. *Cancel:* $\frac{3(x+1)}{6(x+1)}$

ANSWER $\frac{1}{2}$, $x \neq -1$

6. *Cancel:* $\frac{a+b}{2(a+b)}$

ANSWER $\frac{1}{2}$, $a \neq -b$

7. *Cancel:* $\frac{-4x}{8x}$

ANSWER $-\frac{1}{2}$, $x \neq 0$

Medium · the standard question

7 PROBLEMS

1. *Cancel:* $\dfrac{x^2 - 9}{x + 3}$

ANSWER $x - 3,\ x \neq -3$

2. *Cancel:* $\dfrac{x^2 - 4x}{x - 4}$

ANSWER $x,\ x \neq 4$

3. *Cancel:* $\dfrac{2x + 6}{x^2 - 9}$

ANSWER $\dfrac{2}{x - 3},\ x \neq \pm 3$

4. *Cancel:* $\dfrac{a^2 - b^2}{a^2 + 2ab + b^2}$

ANSWER $\dfrac{a - b}{a + b},\ a \neq -b$

5. *Cancel:* $\dfrac{x^2 - 6x + 9}{x^2 - 9}$

ANSWER $\dfrac{x - 3}{x + 3},\ x \neq \pm 3$

6. *Cancel:* $\dfrac{5x - 5y}{x^2 - y^2}$

ANSWER $\dfrac{5}{x + y},\ x \neq \pm y$

7. *Cancel:* $\dfrac{3a - 3b}{b - a}$

ANSWER $-3,\ a \neq b$

Hard · stretch and reason

7 PROBLEMS

1. *Cancel:* $\dfrac{x^3 - x}{x^2 + x}$

ANSWER $x - 1, x \neq 0, -1$

2. *Cancel:* $\dfrac{x^2 - 5x + 6}{x^2 - 4}$

ANSWER $\dfrac{x - 3}{x + 2}, x \neq \pm 2$

3. *Cancel:* $\dfrac{2x^2 + 7x + 3}{x^2 - 9}$

ANSWER $\dfrac{2x + 1}{x - 3}, x \neq \pm 3$

4. *Cancel:* $\dfrac{x^2 - 2x}{x^2 - 4}$

ANSWER $\dfrac{x}{x + 2}, x \neq \pm 2$

5. *Cancel:* $\dfrac{x^4 - 16}{x^2 + 4}$

ANSWER $x^2 - 4$

6. *Cancel:* $\dfrac{m^2 - mn}{n^2 - mn}$

ANSWER $-\dfrac{m}{n}, n \neq 0, m \neq n$

7. *Cancel:* $\dfrac{x^2 + x - 6}{x^2 - 4x + 4}$

ANSWER $\dfrac{x + 3}{x - 2}, x \neq 2$

07 Homework

Homework — 6 problems

Algebra 8, §2, pp. 12–17. State the permissible values with every answer.

1. *Cancel:* $\dfrac{8xy}{12x}$
2. *Cancel:* $\dfrac{x^2 - 25}{x - 5}$
3. *Cancel:* $\dfrac{3x + 12}{x^2 - 16}$
4. *Cancel:* $\dfrac{x^2 + 4x + 4}{x^2 - 4}$
5. *Cancel:* $\dfrac{7a - 7b}{b - a}$
6. Explain in one sentence why $\dfrac{x + 5}{5}$ cannot be cancelled.